September 21, 2022

APS110 – Lecture 6

See hand written notes for drawings.

Density
- <u>Recall</u>: crystalline structure of metals → repeating structure of atoms aligned and bonded into a lattice
  - The order (structure) exists at the atomic level of the metal
- Almost all metals that we deal with are polycrystalline
  - Some exceptions, where we use amorphous metals (not crystalline), or unicrystalline metals
- Scale
  - Order of magnitude is important to think about
  - For a car, you would say it's on the order of a metre
    - It's *not* on the order of 10 metres (which would be the next order up)
  - For the grains of a metal, it's on the order of 10^{-6}m (micron scale) to 10^{-2}m (centimeter scale)
  - For the atom crystal structure, it's obviously on the atomic order (10^{-10}m)
- **Unit cell** – the smallest *convenient* building block of the atomic crystalline structure

Aside: isometric perspective vs. "cabinet-maker's"

Unit Cell:

Face-centered cubic (FCC)
- We define the dimension of the cube as 'a'
  - **a = lattice parameter**
  - We only need to define one dimension of a cube, since all lengths are the same
- **Right-hand positive axes** as convention
  - z-y plane oriented to plane of screen
  - x-axis directs out of the page
- Summary
  - n = 4 (4 atoms inside cube)

"Hard Sphere Model"
- Atoms have defined radii
- Atoms touch at defined distance along the radius
- We typically don't use this model, preferring the reduced sphere model
- You can better visualize here how we get the 1/8 of a sphere at corners of the cube and ½ of a sphere at faces of the cube